

Control Valve Sizing Assignment

Field Devices and Instrumentation

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1 Introduction

The objective of this assignment is to perform a complete control valve sizing exercise for a closed-loop water circulation process. By evaluating the system's pump curve and calculating the necessary pressure drops, a suitable valve size is determined using Valmet Nelprof software. The report compares the current installed hardware with an optimal solution based on flow control theory and justifies the final selection.

2 Process Application Description

The assigned process involves a continuous closed-loop water circulation system. Water at 25°C is pumped from a 0.2 m³ pressure tank (LAA01BB001) and circulated back into the same tank.

Current Installation (Baseline):

- **Pump:** Kolmeks L-32A/2 centrifugal pump equipped with a 105 mm impeller. The motor (OKN-841DN12) runs at 3-phase, 50 Hz, 0.65 kW.
- **Current Valve:** A standard DN 25 ball valve (CF8M/316SS) with a maximum shutoff pressure of 14.7 bar at 38°C or 12 bar at 260°C.
- **Current Actuator:** AUMA electric actuator (SE0R7140-4-LY-L, 220 – 240 V, 50 Hz, 1350 rpm).
- **Flow Requirement:** The target design operating flow rate is 0.85 l/s (3060 kg/h).

3 Calculations and Sizing Methodology

To accurately size the control valve, the pressure differential across the valve must be calculated based on the pump curve and system losses.

3.1 Pump Curve Analysis

According to the manufacturer's performance curve for the Kolmeks L-32A/2 pump with a 105 mm impeller:

- At a flow rate of $q = 0.85 \text{ l/s}$, the pump delivers a head of approximately $H = 14 \text{ m}$.

Converting the operating pump head to pressure (assuming water density $\rho = 1000 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$):

$$p_{\text{pump}} = \rho \cdot g \cdot H = 1000 \cdot 9.81 \cdot 14 = 137\,340 \text{ Pa} \approx 1.37 \text{ bar} \quad (1)$$

3.2 Pressure Loss and Valve Delta P Verwendung von pressure-drop.online (Pumpe → Valve)

Since the fluid is circulated back to the same tank, the static head difference is zero. The entire pump pressure (1.37 bar) is dissipated through piping friction, in-line equipment, and the control valve.

Based on a detailed evaluation of the actual piping isometry, the system comprises:

- **Upstream of Pump:** Approximately 0.88 m of piping including 4 elbows (90°), 1 tee-piece, and 3 manual block valves (LAA01AA006, LAA01AA101, LAA01AA007).
- **Downstream of Pump to Valve:** Approximately 2.59 m of piping including 6 elbows (90°) and several restricting in-line instruments: an ABB ProcessMaster 300 flowmeter (CF202), a SATRON pressure transmitter (CP202), and an SKS temperature sensor (CT202).

The dynamic piping and fitting losses (p_{loss}) generated by these numerous bends and intrusive sensors account for approximately 0.37 bar overall. To achieve the target flow, the available inlet pressure for the control valve is therefore 1.2 barG, and the required outlet pressure to overcome the remaining return-line friction is 0.2 barG. This results in an available pressure drop (Δp_{valve}) of:

$$\Delta p_{valve} = 1.20 \text{ bar} - 0.20 \text{ bar} = 1.00 \text{ bar} \quad (2)$$

According to the Valmet Flow Control Manual, the Process Pressure Ratio Factor (DP_m) characterizes the pipeline and is calculated as the ratio of the valve pressure drop at maximum flow (Δp_m) to the differential pressure across the closed valve (Δp_0). Assuming Δp_0 approximates the pump shutoff head (≈ 1.37 bar):

$$DP_m = \frac{\Delta p_{valve}}{p_{pump}} = \frac{1.00}{1.37} \approx 0.73 \quad (3)$$

A DP_m value of 0.73 indicates excellent valve authority, which is typical for systems with relatively short process piping.

The estimated valve flow coefficient (C_v) requirement can be verified mathematically:

$$C_v = \frac{q}{N_1 \cdot \sqrt{\frac{\Delta p_{valve}}{G_f}}} = \frac{3.06 \text{ m}^3/\text{h}}{0.865 \cdot \sqrt{\frac{1.0}{1.0}}} \approx 3.54 \quad (4)$$

4 Sizing Results (Nelprof)

Using Valmet Nelprof, the calculated process data (0.85 l/s, $\Delta p = 1.0$ bar, $T = 25^\circ\text{C}$) was applied to two different DN 25 valve alternatives to evaluate their installed performance.

The figure displays two screenshots of the Valmet Nelprof software interface, showing the setup for valve sizing. The top screenshot is for the RE-Series valve, and the bottom screenshot is for the M1-Series valve. Both screenshots show the same process data and results, but with different valve specifications.

Top Screenshot (RE-Series):

- Tag Name:** SIZING NAME
- PHASE:** Liquid
- LIQUID:** Water
- DESCRIPTION:**
- COMMENTS:**
- Calculate:** [Button]
- Save:** [Button]
- Filter:** Type search term
- Filter:** Filter
- Filter:** LAA01AA202
- Filter:** Area
- Filter:** Tag
- Filter:** Current Valve
- Filter:** New Valve
- SIZING DATA:** 3rd party valve, Resistor, Actuator
- FLUID PROPERTIES:**
 - Liquid Specific Gravity: 1
 - Critical pressure: 221.2
 - Design pressure min: barG
 - Design pressure max: barG
 - Design temperature min: Celsius
 - Design temperature max: Celsius
- PIPELINE:**
 - Schedule: Std
 - Inlet size: 32 mm
 - Outlet size: 32 mm
 - Insulation type: No insulation
- VALVE:**
 - Type: Segment Valve
 - Pressure class: EN PN40
 - Series: RE
 - Size: Automatic
 - Size reduction: Automatic
 - Recommended valve travel limits: 3, 95
 - Design style: Not applicable
 - Flow direction: Not applicable
 - Seat code: S
 - Trim characteristics: General
 - Trim option: General
- PROCESS DATA:**
 - Liquid flow rate: 0.85 l/s
 - Inlet temperature: 25 Celsius
 - Inlet pressure: 1.2 barG
 - Pressure difference: 1 bar
 - Outlet pressure: 0.2 barG
 - Vapour pressure: 0.0317 barA
 - Dynamic viscosity: 0.0317 cp
- RESULTS:** RE | EN PN40 | SIZE: 25 MM / 1" | REDUCTION: 2 | MAX CAPACITY: 5 CV
- CHARTS:**
- SIGMA VALUES:**

Bottom Screenshot (M1-Series):

- Tag Name:** SIZING NAME
- PHASE:** Liquid
- LIQUID:** Water
- DESCRIPTION:**
- COMMENTS:**
- Calculate:** [Button]
- Save:** [Button]
- Filter:** Type search term
- Filter:** Filter
- Filter:** LAA01AA202
- Filter:** Area
- Filter:** Tag
- Filter:** Current Valve
- Filter:** New Valve
- SIZING DATA:** 3rd party valve, Resistor, Actuator
- FLUID PROPERTIES:**
 - Liquid Specific Gravity: 1
 - Critical pressure: 221.2
 - Design pressure min: barG
 - Design pressure max: barG
 - Design temperature min: Celsius
 - Design temperature max: Celsius
- PIPELINE:**
 - Schedule: Std
 - Inlet size: 32 mm
 - Outlet size: 32 mm
 - Insulation type: No insulation
- VALVE:**
 - Type: Ball Valve
 - Pressure class: EN PN40
 - Series: M1
 - Size: Automatic
 - Size reduction: Automatic
 - Recommended valve travel limits: 5, 90
 - Design style: Seat supported
 - Flow direction: Not applicable
 - Seat code: T
 - Trim characteristics: General
 - Trim option: General
- PROCESS DATA:**
 - Liquid flow rate: 0.85 l/s
 - Inlet temperature: 25 Celsius
 - Inlet pressure: 1.2 barG
 - Pressure difference: 1 bar
 - Outlet pressure: 0.2 barG
 - Vapour pressure: 0.0317 barA
 - Dynamic viscosity: 0.0317 cp
- RESULTS:** M1 | EN PN40 | SIZE: 25 MM / 1" | REDUCTION: 0 | MAX CAPACITY: 105 CV
- CHARTS:**
- SIGMA VALUES:**

Figure 1: Nelprof Setup (Top: RE-Series, Bottom: M1-Series)

4.1 Alternative 1: RE-Series Segmented Ball Valve (Optimal)

The RE-Series segmented valve was selected with a reduced trim (Reduction: 2, Max Capacity: $5 C_v$) to perfectly match the low flow conditions.

- **Required Capacity (C_v):** 3.54
- **Travel / Opening:** 87.8 % (84.7°)
- **Predicted Noise Level:** 42.0 dBA (well below the 80 dBA limit for cavitation damage)
- **Choked Dp:** 1.78 bar (No cavitation occurs as $\Delta p = 1.0$ bar is below this limit)

4.2 Alternative 2: M1-Series Standard Ball Valve (Current Installation)

The standard M1-Series ball valve was evaluated under the exact same process conditions to demonstrate its behavior in a throttling application.

- **Required Capacity (C_v):** 3.55

- **Travel / Opening:** 33.5 % (40.8°)
- **Predicted Noise Level:** <40.0 dBA at operating point (Peak: 53.0 dBA at 18% travel)
- **Choked Dp:** 1.73 bar

5 Graphical Analysis (Nelprof Charts)

The following charts generated by Nelprof visualize the hydrodynamic and control behavior of the selected valves. The optimal RE-series is displayed on top, while the currently installed M1-series is below it for direct comparison.

5.1 Inherent Flow Characteristics

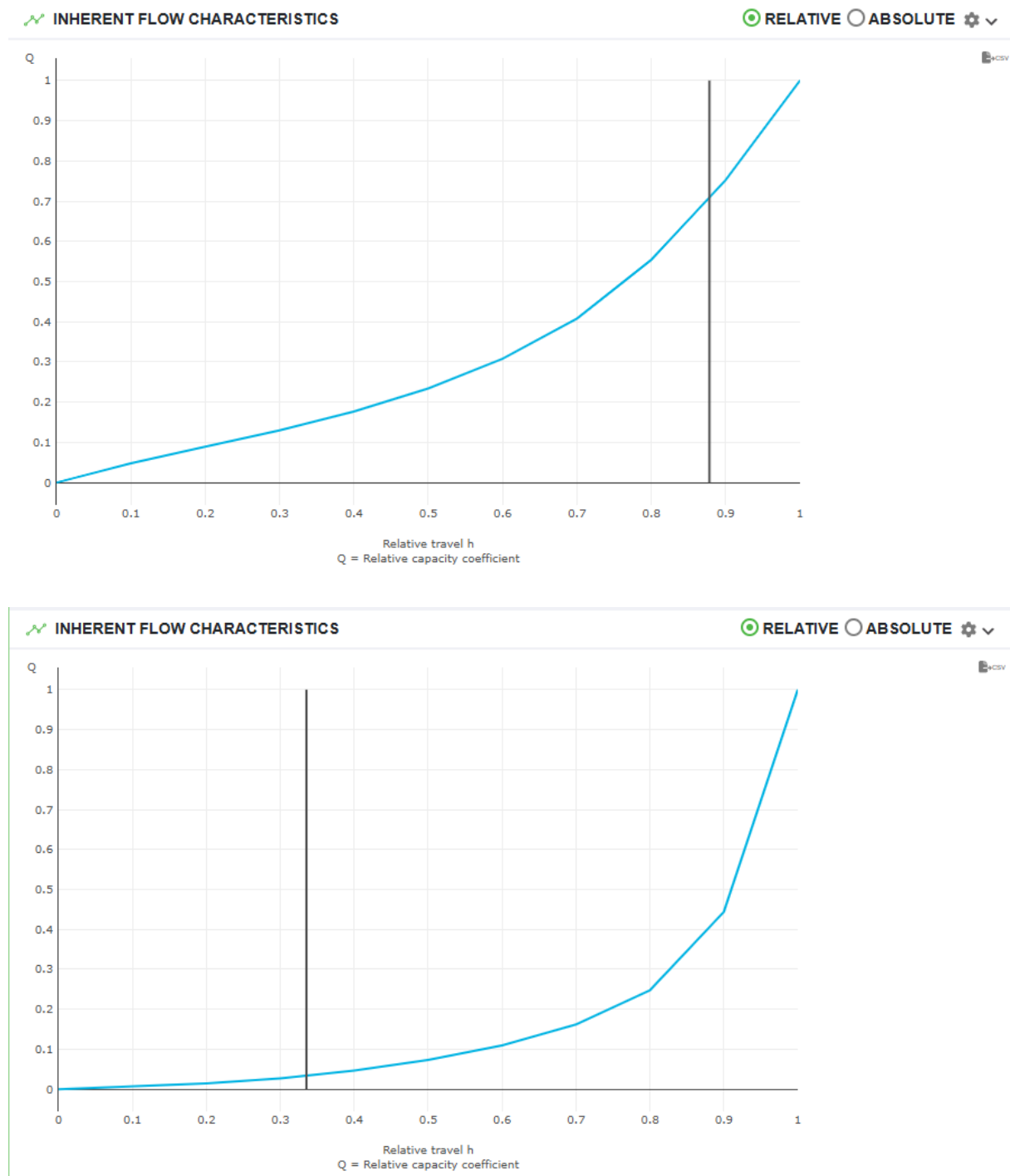


Figure 2: Inherent Flow Characteristics (Top: RE-Series, Bottom: M1-Series)

The inherent flow characteristic displays the theoretical flow capacity (C_v) of the valve as a function of relative travel, assuming a constant pressure drop across the valve. The simulated data reveals a highly desirable equal-percentage profile for the RE-series segment valve (top). At 50% travel, its relative flow capacity is restricted to 23.3%, progressively opening to 55.4% at 80% travel. This specific exponential curve design effectively compensates for dynamic system pressure losses at higher flow rates. In contrast, the standard M1 ball valve (bottom) has a massive total capacity of $105.0 C_v$. Because it is so heavily oversized for the required $3.55 C_v$,

its inherent curve relative to the process rises extremely steeply right from the beginning, which makes fine throttling inherently difficult.

5.2 Installed Flow Characteristics

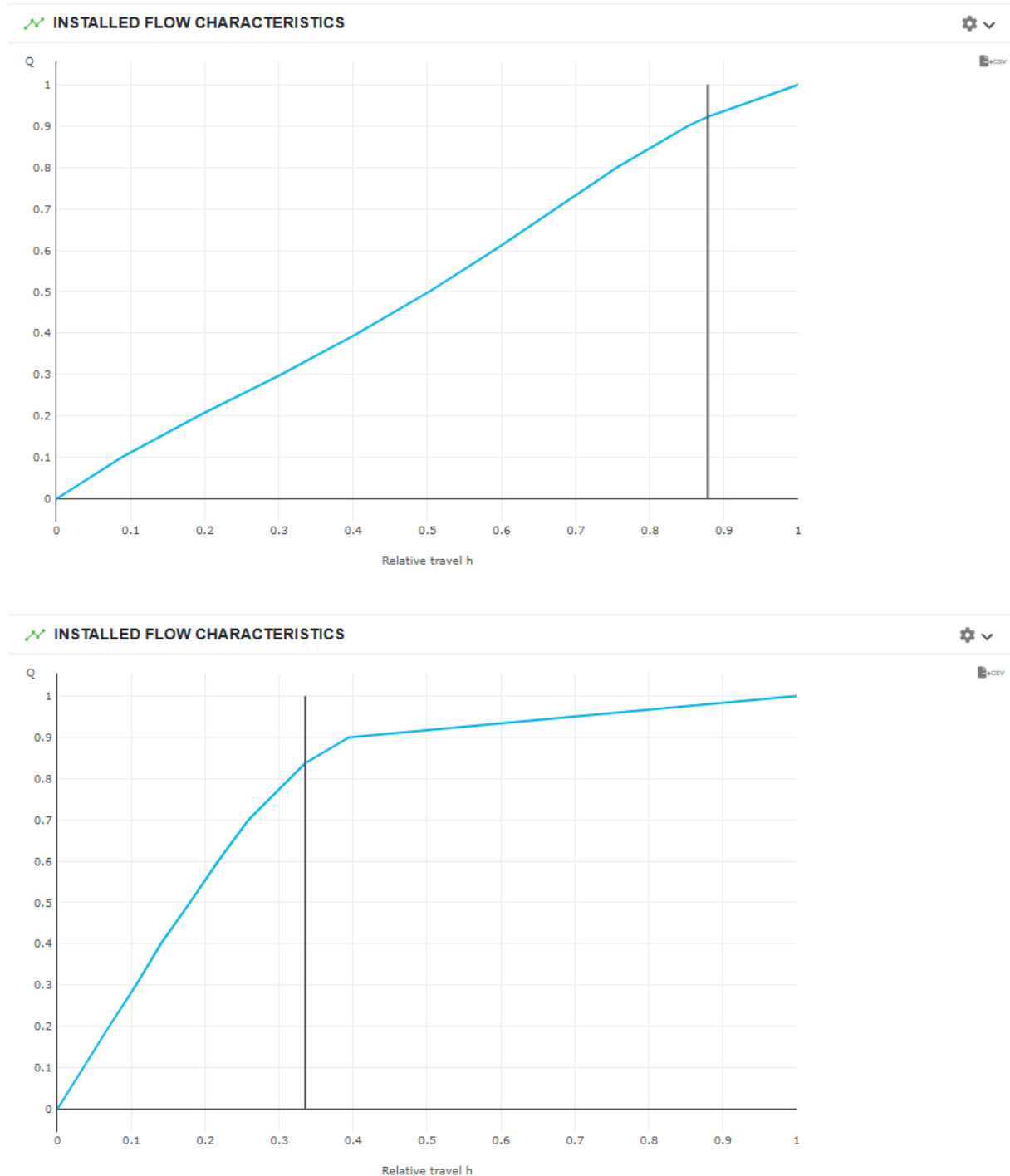


Figure 3: Installed Flow Characteristics (Top: RE-Series, Bottom: M1-Series)

Unlike the inherent curve, the installed characteristic accounts for the actual dynamic pressure losses in the connected piping. A properly sized control valve should yield a nearly linear installed flow characteristic to ensure predictable process behavior. The simulation data proves that the selected RE valve (top) achieves exactly this: the relationship between relative travel and flow

is almost perfectly linear (0.500 travel yields ≈ 0.500 flow; 0.800 travel yields ≈ 0.850 flow). In stark contrast, the oversized M1 valve (bottom) reaches 80% of the installed flow at merely 31.3% travel, and 90% at 39.4% travel. This severely compresses the entire usable control range into the first third of the valve opening, rendering the remaining 60% of its mechanical travel useless for this process and eliminating any high-resolution control.

5.3 Installed Gain

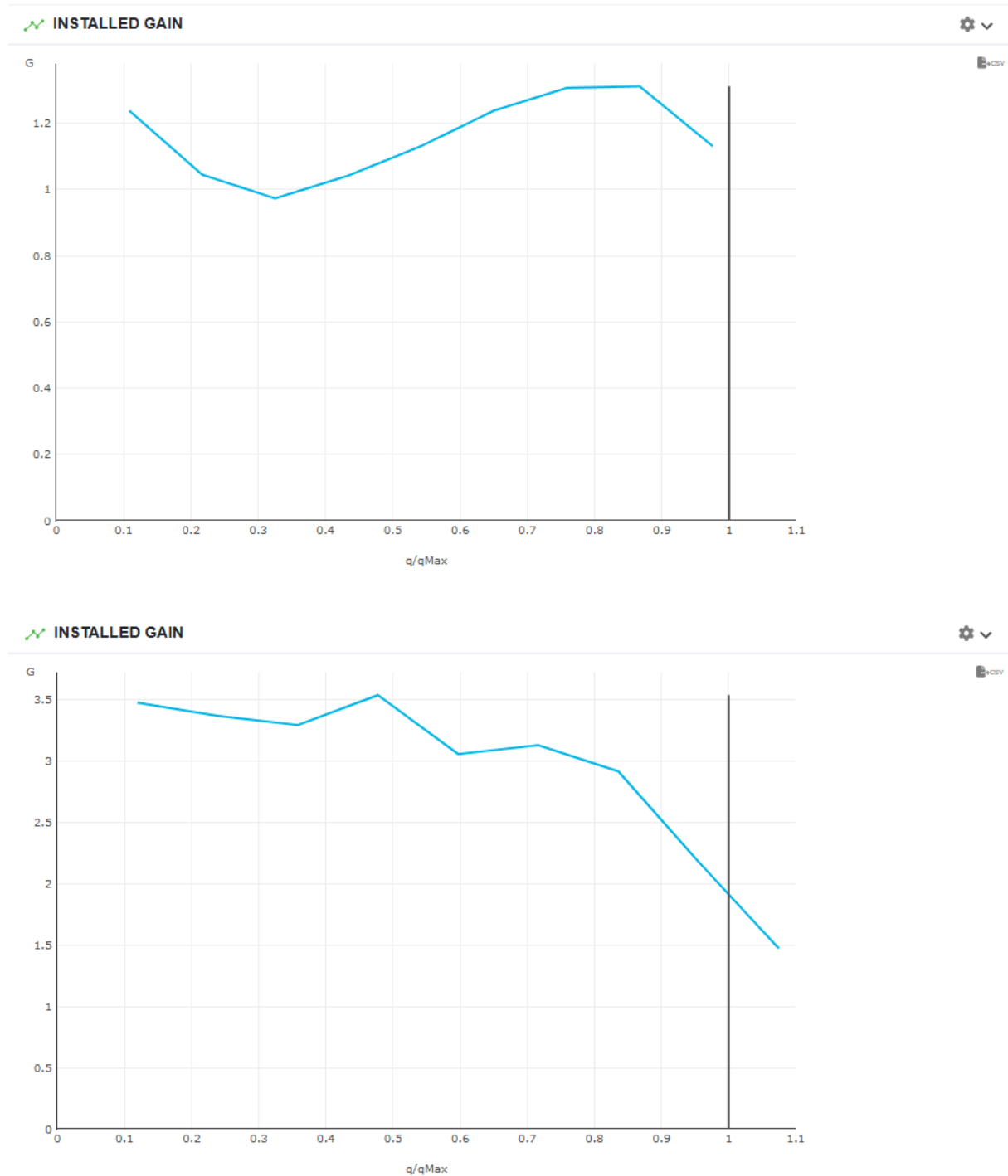


Figure 4: Installed Gain Curve (Top: RE-Series, Bottom: M1-Series)

The installed gain ($G = dQ_p/dh$) represents the sensitivity of the flow rate to changes in valve position and is the most critical parameter for process stability. According to the Valmet Flow Control Manual, the installed gain must ideally remain between 0.5 and 2.0. The simulated RE-series valve (top) performs flawlessly within these limits: the gain remains highly stable between a minimum of 0.97 and a maximum of 1.31.

Conversely, the M1-series (bottom) exhibits an excessively high installed gain, hovering between 3.2 and 3.53 in the primary control range. A gain of 3.5 means that even a microscopic 1% change in valve position causes a massive 3.5% change in flow. This severe violation of the flow control guidelines inevitably leads to control loop oscillation (hunting) and unstable PID controller performance.

5.4 Installed Pressure Level

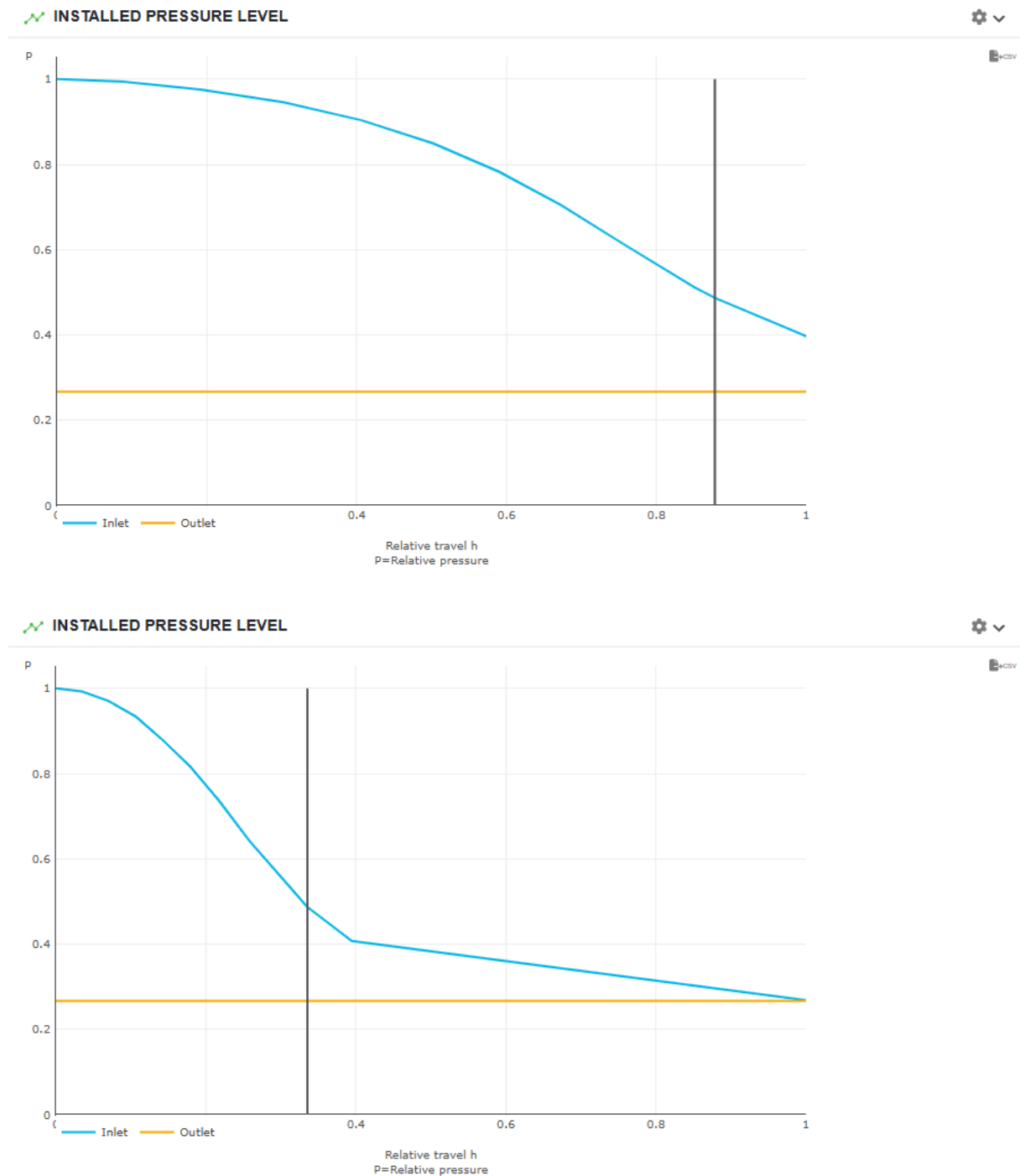


Figure 5: Installed Pressure Level (Top: RE-Series, Bottom: M1-Series)

This graph illustrates the variation of the upstream and downstream pressures as the flow rate changes. The simulation confirms the theoretical pipeline calculations: as flow increases to maximum capacity, dynamic pipe friction slightly decreases the relative pressure available directly at the valve inlet (from 1.000 to 0.400 for the RE valve, and 1.000 to 0.268 for the M1 valve). Both valves handle the differential pressure window effectively without dropping below the vapor pressure boundary (0.0317 barA), ensuring that no flashing occurs in the system.

5.5 Installed Noise

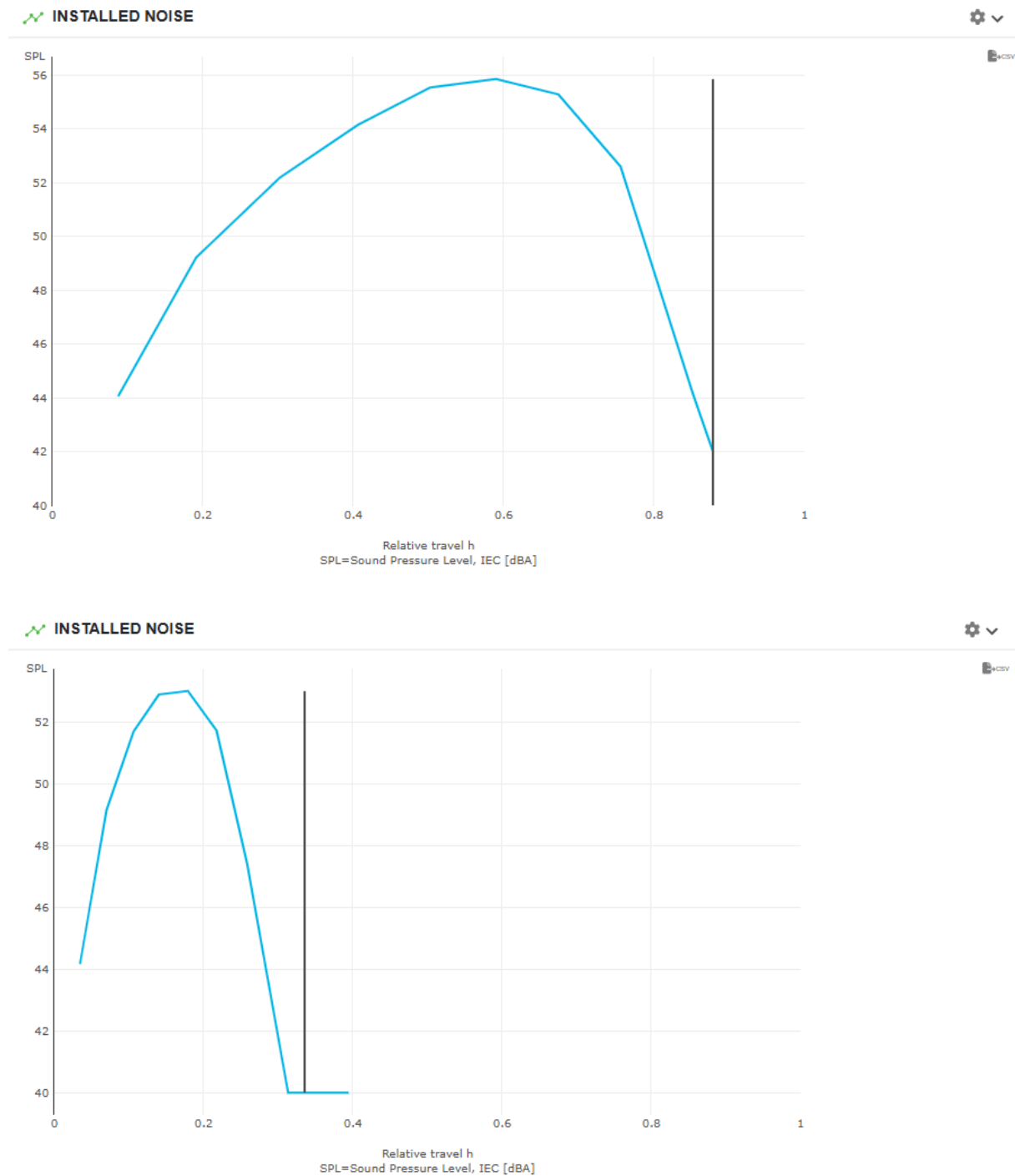


Figure 6: Installed Hydrodynamic Noise (Top: RE-Series, Bottom: M1-Series)

Hydrodynamic noise serves as the primary indicator of cavitation intensity and mechanical wear. The predicted sound pressure level for the RE-series peaks at 56.0 dBA at roughly 60% travel. The M1-series experiences a brief noise peak of roughly 53 dBA at 18% travel, dropping to 40 dBA at its operating point. According to established flow control standards, the critical noise limit for cavitation damage in this pipe schedule is 80 dBA. At these low acoustic levels, both valves operate silently and are immune to cavitation-induced mechanical damage or erosion.

6 Comparative Evaluation and Conclusions

6.1 Valve Performance Comparison

The current installation utilizes a standard ball valve (M1-series). While standard ball valves offer excellent, tight shut-off capabilities and high maximum flow capacities, their inherent flow characteristic is severely flawed for accurate throttling duties. As mathematically proven by the Nelprof simulations, the massive $105.0 C_v$ capacity of the M1 valve forces the process to operate within a highly compressed travel range (reaching 90% flow at only 39.4% opening). Furthermore, it exhibits an aggressively high installed gain peaking at 3.53. This mismatch inevitably leads to control loop oscillation (hunting) and poor flow stability.

In contrast, the RE-series segmented ball valve (utilizing a reduction 2 trim) provides the optimal solution for this specific circulation loop. By utilizing an equal-percentage inherent flow characteristic, the valve completely neutralizes the dynamic friction losses of the piping network, resulting in a nearly perfectly linear installed flow curve with excellent resolution. Most importantly, the installed gain is constrained within the ideal textbook boundaries of 0.97 and 1.31. This eliminates the severe high-gain instability typically associated with standard on/off ball valves when forced into throttling duties.

6.2 Actuator Comparison

The currently installed AUMA electric actuator is highly robust. However, electric actuators of this multi-turn or slow quarter-turn type are traditionally better suited for ON/OFF isolation duties where speed is not a primary concern. An optimal control setup for the RE-series valve would utilize a pneumatic actuator (e.g., a diaphragm or rack-and-pinion design). Pneumatic actuators deliver high-speed response times, seamless integration with smart positioners (e.g., Neles ND9000), and a reliable mechanical spring-return for defined fail-safe states, significantly improving the dynamic responsiveness of the control loop.

6.3 Final Verdict

Based on the process parameters, manual piping calculations, and Nelprof software analysis, the **RE-series segmented ball valve (DN 25)** paired with a **pneumatic actuator and smart positioner** is the optimal selection for this application. It entirely resolves the severe control limitations of the currently installed standard ball valve by providing a stable installed gain and excellent resolution, ensuring precise, real-time flow modulation without the risk of control loop hunting.

References

- [1] JAMK University of Applied Sciences. (2024). *Valve structures and control valves* [Learning material]. Field Devices and Instrumentation.
- [2] Kolmeko. (2012). *Pump performance curve: L-32A/2 (105mm impeller)*.
- [3] Valmet. (2022). *Flow control manual* (6th ed.). Neles Finland Oy.